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Stoffregen's (this issue) target article reminds us of many things about affordances and events, but it also argues for points that cannot be maintained. In particular, Stoffregen argues that animals do not perceive events. This result, if it turned out to be true, would create widespread consequences for ecological psychology. I argue that Stoffregen is incorrect: We do perceive events. To see that this is the case, however, requires a new conceptualization of events in which events are not merely changes in the physical surrounds of an animal; instead, events are taken to be changes in the layout of affordances of the animal–environment system. Some consequences of this conception of affordances are sketched.

Thomas Stoffregen's "Affordances and Events" (this issue) makes many points that are forgotten all too often—if they are realized at all—by adherents to the ecological perspective in psychology. He is to be applauded for this. But he compiles these points to make a very strong and very sweeping claim about the validity of a broad swath of research that is done by ecological psychologists. In particular, he argues for the following conclusion:

More generally, I have suggested that events may not be perceived; it may be that only affordances are perceived. If so, this would raise serious questions about the utility of research on event perception in the context of the ecological approach. (p. 23)

Strong words, indeed: If this is right, much of the work that has been published in this journal over the years would be relegated to the scrap heap. In what follows, I show that ecologically oriented psychologists can accept most of Stoffregen's claims but still study events and event perception with good conscience. Doing so, however, requires a significant conceptual, and perhaps empirical, restructuring.

The most important of Stoffregen's (target article, this issue) points, and the ones that hold up to scrutiny, concern differences between affordances and events as they are normally conceived by experimentalists. A brief rundown of these points is in order:

1. Events and affordances are not the same. Affordances are "relations between an animal and its environment that have consequences for behavior" (p. 7). Events are normally conceived as "static and dynamic properties of objects and surfaces defined without reference to behavior and not scaled relative to action-relevant properties of the animal" (p. 15).
2. As a matter of practice, events are measured in standard, extrinsic units. When they have been measured at all, affordances are measured differently, for example, using unit-free pi-numbers (e.g., by Warren, 1984).
3. Events are fully objective properties of the physical world. Affordances are not fully objective in that they make sense only in the context of an animal-environment system.¹

Combine these points concerning the differences between affordances and events as normally conceived with the ecological truism that animals perceive affordances (Gibson, 1979/1986; Sanders, 1997), and it follows fairly directly that events are not perceived at all.² This, in turn, leads Stoffregen to question the value of event perception research. And rightly so, assuming that events really are as Stoffregen defines them, and as they have been conceived by event perception researchers, as objective properties of physical objects and surfaces, properly measured in kilograms, reflectances, and the like.

However, it seems obvious that we can perceive events. We seem to do so both as scientists in the lab and as regular folk who stop to watch the sun setting as we walk home from an after-work pint. Given this, the ecological perspective must be able to either account for event perception or explain why it (merely) seems so obvious that we perceive events like sunsets, not to mention the moon landing and Mark McGwire's 62nd home run. I suggest that ecological psychology can explain event perception. Despite Stoffregen's (target article, this issue) arguments, we can save event perception research and show how ecological psychologists can account for event perception, but doing so will require altering our conception of events. The basis for such a reconception is another ecological truism: "Affordances are ontologically prior to events" (p. 8; see also Chemero, 1999; Reed, 1996; Sanders,

¹This is not to say that affordances are fully subjective or depend on any particular animal. As Gibson (1979/1986) put it, "an affordance is neither an objective property nor a subjective property; or it is both if you like. An affordance cuts across the dichotomy of subjective-objective and helps us understand its inadequacy" (p. 129). However, affordances do depend on the existence of types of animals. The point is that if no perceivers ever lived, events would occur, but there would be no affordances (see Chemero, 1999).

²Stoffregen (target article, this issue) backs this up by arguing that there are no obvious evolutionary benefits to perceiving events. I discuss this more later.

1997). If we suppose that this is the case, we are led to try to derive perceived events from affordances. Doing so underwrites an understanding of events much different from the physics-based one Stoffregen ascribes to event perception researchers. *Events are changes in the layout of affordances of the animal–environment system.*³

Conceiving events this way seems perfectly natural from the ecological perspective and lends itself well to quantification. To see this, start by considering the climbability judgment task described in Warren (1984). This now-classic experiment, in which participants were asked to judge visually whether stairs of varying heights afforded climbing (from a standing position, without “boosting” using the hands and arms), found that there was a constant ratio between the participant’s leg length and the height of the tallest riser the participant judged to be climbable. Because both the leg length and stair riser height are measured in centimeters, those units can be divided out, leaving a dimensionless, pi-number. Warren found that participants judged that stairs were climbable as long as

$$\pi = r/l \leq .88$$

where r is the stair riser height and l is the participant’s leg length. Notice that this is a quantification of an aspect of the environment relative to properties of the participant and the participant’s potential behaviors, which makes it an affordance by Stoffregen’s (target article, this issue) lights. Also, $\pi = .88$ is a critical point, above which the affordance for stair climbing disappears. Now imagine the following experiment: Place the participants before a stair with a smoothly moving riser. Start with the riser height at 0 cm, then slowly move the riser up, asking the participant to press a button when the stair is no longer climbable. At some point during the riser’s ascension, it will cross the critical point and the stair will no longer afford climbing. The participant who presses the button has perceived a change in the layout of affordances, that is, an event according to the definition offered here.

To my knowledge, the experiment just presented has never been performed, perhaps partly because the task described is not one that humans normally have to do. A possible alternate experiment, one that relies on abilities in the usual human arsenal, would be an extension of Burton’s (1992, 1993) work on gap crossing. Burton asked his participants to judge both visually and nonvisually (using canes) whether gaps between platforms were crossable. Burton’s experiments could be modified similarly to the proposed modifications to Warren’s (1984) experiments, so that participants are asked to determine the point at which the gap between moving platforms becomes uncrossable. Asking participants to judge when a gap between moving platforms is no longer crossable has analogues in normal experience—for example, when one decides whether one will be able to step onto a boat

³Note that there are all kinds of events: everyday ecological events, quantum mechanical events, celestial events, and so on.

as it pulls away from the dock. Just as with the imagined climbability experiment earlier, a participant who decides that the gap has gone from being crossable to being uncrossable (or from being crossable by stepping to being crossable by jumping to being uncrossable) has perceived an event.

Notice that in perceiving events of this type, participants have not perceived merely physical changes in their environments. Instead, they have perceived action-related, personally scaled changes in their surroundings. In the imagined experiments, participants would see these values cross critical boundaries such that affordances that were present are no longer present. This appearance or disappearance of affordances is not a necessary condition on events: Events as I am defining them are any changes in the values of animal- and action-related variables, whether a critical point is crossed. The crossing of a critical point is necessary experimentally to make sure that the participant is perceiving changes in affordances and not just changes in his or her physical surroundings.

In situations in which affordances are quantified as pi-numbers (as in Warren, 1984), the magnitude of events can be quantified as differences in pi-numbers over time:

$$\text{Magnitude}_{\text{event}} = \Delta\pi = \pi_{t_1} - \pi_{t_0}$$

Different types of events will have characteristic rates, quantified as the first temporal differential of the pi-numbers, $d\pi/dt$, with larger differentials for gap-crossing affordances (when the gap in question is between the dock and a boat pulling away from it), and smaller differentials for ripening tomatoes (which can take weeks), and still smaller differentials for stair-climbing affordances, which change only as fast as the leg length of human children.⁴

Along with its obvious intuitive appeal, there are additional benefits that are gained by conceiving events as changes in the layout of affordances in the animal-environment system. First, events, like affordances, are not simply objective properties of the physical world; they make sense only in the context of the affordances of the animal-in-the-world. Second, conceiving perceived events in this way shows a way for event perception researchers to get past Stoffregen's (target article, this issue) main challenge to them: "My analysis poses challenges for researchers who study event perception within the ecological approach to perception and action. The challenge is to derive from ecological theory a motivation for the perception of events." (p. 21)

He argues that there is good evolutionary reason to perceive affordances, which are immediate opportunities for behavior, but there is no good reason for event-perceiving abilities to have evolved. This is, no doubt, true of events conceived of as changes in the physical features of objects or surfaces. But it is false of events as we have reconceived them here. Events (*qua* changes in the layout of affordances)

⁴Or as fast as our skills deteriorate in old age.

should be perceived for all the same reasons that affordances themselves should be perceived. Perceiving this type of event provides obvious evolutionary advantages. Surely it is worthwhile to know that something once did, but no longer does, afford eating, for it may afford eating again later. Thus, a new conception of events undermines Stoffregen's most convincing argument that events are not perceived.

There are three additional consequences that come with redefining events as we have here, one or more of which may strike some as unappealing. First, some will claim that for every change in the layout of affordances of an animal–environment system, there will be some change in the properties of the objects of surfaces of the relevant section of the physical world. That is, they will argue events as we have redefined them here just are physical events. The reply to this sort of reductionist is that it may be that every time an experimental participant reports perceiving an event, there are changes in the physical layout, but these physical changes are simply not identical to perceived events by any reasonable understanding of identity. Analogously, an eraser is a mere chunk of rubber in a world with no pencils, so the eraser is not identical to any chunk of rubber.

A second consequence that some will find disheartening is that it may be the case that events are not perceived directly.⁵ If events are changes in the layout of affordances, which are perceived directly, it may require an inference to perceive that an event has occurred. In particular, it may require attempting to match the pi-number of some region at time t with the pi-number of the same region at time $t - 1$ and realizing that they do not match. That is, perceiving events may require comparing the current affordance with a remembered affordance. Perceiving events thus would be indirect. In response to this, it is worth noting that no psychologist claims that we are incapable of indirect perception: We read books and watch television, after all. Perhaps events are among those things that are perceived only indirectly, or perhaps we actually do perceive changes among affordances directly, without performing such a comparison—remember that evolution is much smarter than we are.

Finally, we must note that event perception research that has taken events to be changes in the physical layout must be reinterpreted in terms of this new understanding of events. Some apparently stable results, no doubt, will not survive reinterpretation. This is a very serious consequence, but it pales in comparison to Stoffregen's (target article, this issue) suggestions that we do not perceive events at all. It is testament to the sharpness of Stoffregen's article that, even though reconceiving events as changes in the layout of affordances blunts his critiques significantly, avoiding their most severe consequences, some apparently good empirical work may not stand up to them. But Stoffregen's desired proscription of event perception research, his boldest and most sweeping conclusion, simply does not stand.

⁵Certainly some events are perceived indirectly, as this morning's puddles can cause us to perceive last night's rainstorm. The worry here is that all events might be perceived indirectly.

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